PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

***ExamLens**: A syllabus and previous-year question paper analyzer*

Student / Team Information

Team Name: *Team #*	*FRACTAL VERSE (DSCPP-III-2026-T309)*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Nauriyal, madhav – 2510011324* *madhavnauriyal2401@com* 
Team member 2 *(Last Name, name: student ID: email, picture):*	*Sharma, Aarav – 2510012730* *Aaravsharma2642@gmail.com* 
Team member 3 *(Last Name, name: student ID: email, picture):*	*Ishant – 2510370272* *ishantbahukhandi.8@gmail.com* 
Team member 4 *(Last Name, name: student ID: email, picture):*	*Sinha, Sasmit – 2510012002* *sasmitsinha2009@outlook.com* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Prior to each exam, our group performs the same routine. A collection of previous papers is shared, someone inquires about the important units, and four or five of us gather with a notebook to tally. We followed this process for TMA 316 using the four papers we received, spanning from October 2022 to December 2023[6].On a subsequent evening, we realized that Relations and Lattices had been covered in all four papers, while Recurrence Relations appeared in just one. It was an evening dedicated to four papers on a single topic.ExamLens operates in an automated manner during the evening. Provide it with a syllabus and a selection of five to ten years of examination papers, and it will identify which subjects are frequently revisited.what specific questions were posed for each subject and in which year, how points are distributed among different sections, which subjects are gaining or losing importance, and where to focus your efforts when there are three days remaining.It won't indicate that a question is on the way. It will mention that a subject has significant importance for study due to its frequency of occurrence, how recent it is, the number of points it was worth, and how often the same question was asked again.That statement should be placed in the top section of the report to prevent anyone from interpreting it as a forecast.This is present in both classes without extending. Asking the same questions again requires a hash map based on the revised question text. Associating a question with a syllabus subject involves string matching against a trie of topic keywords, and that trie also works for autocomplete. The top five priorities from several hundred scored subjects are managed in a heap[1],[2].Trends are a kind of data. In C++, each analysis is a specific type of an abstract Analyzer. The first phase is a tool that works through the command line, taking text files in and giving a report out. During the final stage or afterward, we deal with PDF extraction, matching rephrased questions, and anything related to natural language processing.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

At this moment, there are three methods to achieve this, but none are effective. You can tally manually or use a WhatsApp poll, which takes time, relies on the person who counted, and has to be repeated for the next group.Coaching-center "key questions" PDF files represent someone's views from a year or two ago without any numbers included. There are chatbots that allow you to input a document and inquire about its significance. The responses seem good, but they are not based on any data, they cannot process ten documents at once, and they provide different responses every time you ask.Analysis of previous year questions on a large scale is available for JEE and UPSC. These platforms display counts for each topic, but they operate on carefully selected databases for a specific syllabus.No one has created this for the specific subject codes of a university, and there is no tool that allows students to input their personal syllabus and assignments. ExamLens provides this functionality and operates without an internet connection on a lab computer.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

One command is examlens analyze –out report. md, which generates a report. This report includes each question categorized by its unit and topic, along with the corresponding year listed next to it. It also shows total counts for each topic, including how many questions there are, how many years those questions appeared over the total years, and breaks down the questions by their mark values of 2, 5, and 10. The report highlights repeated questions along with the years they appeared, identifies the most and least frequently asked questions, provides a trend for each topic classified as consistent, increasing, decreasing, or occasional, offers overall totals by unit, and assigns a priority score ranging from 0 to 100, classified as VERY HIGH, HIGH, MEDIUM, or LOW. A second command, examlens suggest, helps finish the topic names based on the syllabus provided.Four steps, each with a check we must complete before progressing.

- *Stage one, Parsing. Lock the text input style. The parser transforms syllabuses and papers into objects like Syllabus, Unit, Topic, Question, and QuestionPaper, while dismissing incorrect lines. This step is finished when all four TMA 316 papers are parsed, and every question is printed with its respective year, number, and marks.*
- *Stage two, Storage and matching. The core of C includes hash maps that connect question text to counts and years, topics to questions, and years to questions; a trie is used for syllabus keywords; KMP tags each question with a relevant topic and unit. This stage is completed when each TMA 316 question is categorized under its topic along with a year, duplicate entries are identified, and suggestions are made.*
- *Step 4, Evaluating and ranking. A base class called Analyzer with subclasses for Frequency, Unit-Trend, Marks, and Importance, using a min-heap for the best results, organizing data for trends, and allowing custom operations on Question. It's complete when the TMA 316 report with levels fits our manual count.*
- *Step 5, Producing and strengthening. A ReportWriter for the terminal, including report. md and CSV, handling errors for faulty files, a Python script that converts a PDF document into our text format, a different subject, timing, report, demonstration.*

Extra: a SimilarityAnalyzer that clusters rephrased questions based on token overlap and edit distance.

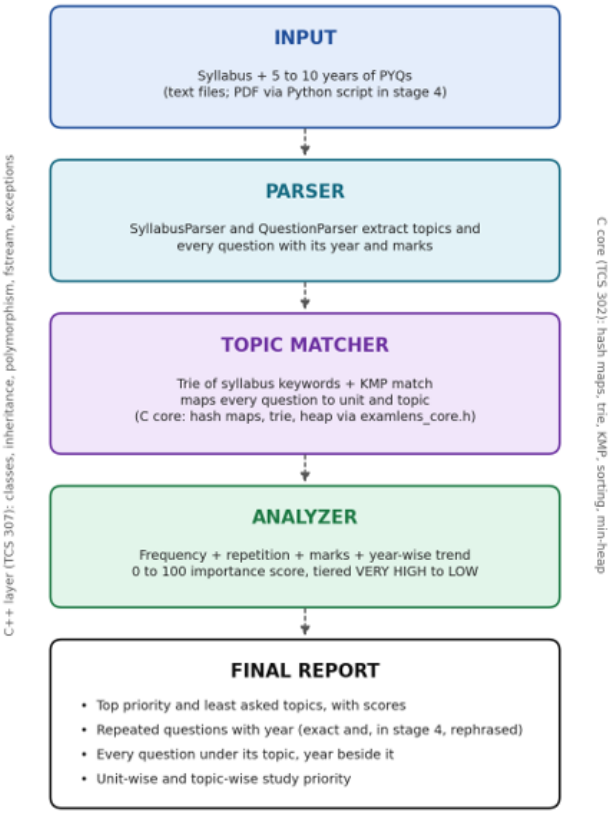
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

Team Task Distribution
Madhav: Develops the main C data structures like hash maps, tries, and min-heaps.[1], [2].
Aarav: Creates the structure for the C++ Object-Oriented programming and connects different modules[5].
Ishant: Manages reading text files and sets up the first TMA 316 testing dataset[6].
Sasmit: Produces the final reports and checks the results against manually counted documents.
Core Setup Inputs
To concentrate on the main software development ideas we are learning this year, we are using C for data structures and C++ for creating classes[2]. We will compile the code using g++ on the computers in the lab and use GitHub for managing different versions. To keep it simple, our input will just be basic plain text files for both the syllabus and previous question papers[6].
Applying Data Structures
Hash Maps: To quickly count the occurrences of questions and note the years, with the goal of reaching an average O(1) lookup time[3].
Tries: To search for and match keywords from the syllabus effectively.
KMP Algorithm: To find matches between exam questions and syllabus topics in O(n + m) time[4].
Min-Heaps: To quickly access the most important topics without needing to sort the entire list each time, working in O(n log k) time[3].
Scoring OOP Logic
Each topic receives a priority score from 0 to 100 based on how many times it appears, its recent exams, and the marks. The mathematical multipliers are stored in a different text file so we can modify them easily without having to recompile the code.
To actively practice our semester's Object-Oriented Programming syllabus, we will use inheritance with a main Parser class that divides into syllabus and question parsers, virtual functions for our analysis classes using standard design patterns, and basic operator overloading to print question data easily. We will verify the final code by comparing its results to those of manually tallied past TMA 316 papers[6].

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Top to bottom is the order data moves. Parser, Topic Matcher, Analyzer and Report are C++ classes. Underneath them, in C behind examlens_core.h, are the hash maps, trie, KMP and min-heap.

Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

- *A GitHub repository containing the C core, the C++ layer, a Makefile, a README that includes instructions for building on macOS and Linux, and the PDF script.*
- *The examlens executable that includes analyze and suggest features, along with the documented input format.*
- *The TMA 316 dataset, its syllabus, four related papers, and the resulting report.*
- *A dataset and report from another subject.*
- *A project report that links every structure and class to the relevant syllabus unit, provides the importance formula and its weights, and includes a complexity chart for insert, lookup, tag, and top-k operations.*
- *A demonstration lasting five minutes. Execute it on TMA 316, display the priority list, type bin into the suggest feature, modify one paper, and run it again.*

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

Phase 1 documents are entered into our text style. PDF extraction is a part of a stage-4 script, while scanning and OCR of papers are not included. The syllabus includes unit titles and topic headings. Each question line has a number and a score. A topic is part of a single unit. There are five to ten papers for each subject along with several hundred questions that can be stored in memory. Questions are written in English. The output reflects past priorities and is not a forecast. Users are familiar with navigating a terminal.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

- *[1] E. Horowitz and S. Sahni, Fundamentals of Data Structures, Galgotia.*
- *[2] A. M. Tenenbaum, Y. Langsam and M. J. Augenstein, Data Structures Using C and C++, PHI.*
- *[3] T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, Introduction to Algorithms, 3rd ed., MIT Press. Chapters 11 (hash tables), 6 (heaps) and 32 (string matching).*
- *[4] D. E. Knuth, J. H. Morris and V. R. Pratt, "Fast pattern matching in strings", SIAM Journal on Computing, vol. 6, no. 2, 1977.*
- *[5] E. Gamma, R. Helm, R. Johnson and J. Vlissides, Design Patterns, Addison-Wesley, 1994. Strategy pattern.*
- *[6] GEU syllabus documents for TCS 302, TCS 307 and TMA 316; TMA 316 question papers, Oct 2022, Jan 2023, Oct 2023, Dec 2023.*